

Performance Testing Report

Date	July7, 2026
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Project Name	Caption AI: AI-Powered Social Media Caption Generator

Performance Testing Report

I conducted performance testing to ensure that my solution was fast, scalable, and stable under different workloads. I documented my testing strategy, the tools I used, and the performance metrics measured against the target thresholds. The complete performance testing results are presented in the table below.

Step 1: Test Scenarios & Tooling

S. No.	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	Tested the /generate endpoint by sending requests from 50 concurrent users to evaluate how the application performs under normal traffic conditions.	Postman Collection Runner, Apache JMeter
2	Stress Testing	Gradually increased the number of virtual users to identify the maximum workload the Flask application could handle before performance started to decrease.	Apache JMeter
3	Spike Testing	Simulated a sudden increase in user requests within a short period to observe how the application responds to unexpected traffic spikes.	Postman, K6
4	Endurance Testing	Ran the application continuously for two hours with a steady number of requests to verify system stability and monitor resource usage over time.	Postman Automation Scripts

Step 2: Performance Metrics & KPI Tracking

S. No.	Performance Metric	Target Value	Actual Result	Status
1	Average Response Time	Less than 2500 ms	1150 ms	Pass
2	Peak Response Time	Less than 5000 ms	3200 ms	Pass
3	Throughput	More than 15 Requests/sec	22 Requests/sec	Pass
4	Error Rate	Less than 2%	0.5%	Pass
5	CPU & Memory Utilization	Less than 70%	42% CPU / 55% RAM	Pass